Part 1: Linear Algebra

Q1. $A=\begin{bmatrix}3 & 4\\ 2 & 1\end{bmatrix}$ $B=\begin{bmatrix}1 & 2\\ 0 & -1\end{bmatrix}$

$AB=\begin{bmatrix}3 & 4\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2\\ 0 & -1\end{bmatrix}=\begin{bmatrix}3 & 2\\ 2 & 3\end{bmatrix}$ $(A+B)^T=\begin{bmatrix}4 & 6\\ 2 & 0\end{bmatrix}^T=\begin{bmatrix}4 & 2\\ 6 & 0\end{bmatrix}$

$BA=\begin{bmatrix}1 & 2\\ 0 & -1\end{bmatrix}\begin{bmatrix}3 & 4\\ 2 & 1\end{bmatrix}=\begin{bmatrix}7 & 6\\ -2 & -1\end{bmatrix}$ $AB^T=\begin{bmatrix}3 & 4\\ 2 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 2 & -1\end{bmatrix}=\begin{bmatrix}11 & -4\\ 4 & 1\end{bmatrix}$

trace $(A)=3+1=4$

trace $(AB)=3+3=6$

Q2. $M=\begin{bmatrix}2 & -1\\ -1 & 2\end{bmatrix}$ $\rightarrow \det(M-\lambda I)=0$

$M-\lambda I=\begin{bmatrix}2-\lambda & -1\\ -1 & 2-\lambda\end{bmatrix}$ $\rightarrow \det(M-\lambda I)=(2-\lambda)^2-1=\lambda^2-4\lambda+3=(\lambda-3)(\lambda-1)=0$

$\rightarrow \lambda_1=1,\ \lambda_2=3$

① $\lambda_1=1$

$(M-\lambda I)v=\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}v=0$, Let $v=\begin{bmatrix}v_1\\ v_2\end{bmatrix}$ $\rightarrow$ $\left.\begin{matrix}v_1-v_2=0\\ -v_1+v_2=0\end{matrix}\right\}\Rightarrow v=\begin{bmatrix}1\\ 1\end{bmatrix}$

② $\lambda_2=3$

$(M-\lambda I)v=\begin{bmatrix}-1 & -1\\ -1 & -1\end{bmatrix}v=0$, Let $v=\begin{bmatrix}v_1\\ v_2\end{bmatrix}$ $\rightarrow -v_1-v_2=0$, $v=\begin{bmatrix}1\\ -1\end{bmatrix}$

$\Rightarrow M\begin{bmatrix}1\\ 1\end{bmatrix}=\begin{bmatrix}1\\ 1\end{bmatrix}=1\begin{bmatrix}1\\ 1\end{bmatrix}$, $M\begin{bmatrix}1\\ -1\end{bmatrix}=\begin{bmatrix}3\\ -3\end{bmatrix}=3\begin{bmatrix}1\\ -1\end{bmatrix}$

Q3. By definition, $\text{trace}(AB) = \sum_{\hat{n}} \sum_{\bar{j}} A_{\hat{n}\bar{j}} B_{\bar{j}\hat{n}}$ and $\text{trace}(BA) = \sum_{\hat{n}} \sum_{\bar{j}} B_{\bar{n}\bar{j}} A_{\bar{j}\hat{n}}$

$$= \sum_{\bar{j}} \sum_{\hat{n}} A_{\hat{n}\bar{j}} B_{\bar{j}\hat{n}}$$
$$= \sum_{\bar{j}} \sum_{\hat{n}} B_{\bar{j}\hat{n}} A_{\hat{n}\bar{j}}$$

since $\sum_{\bar{j}} \sum_{\hat{n}} B_{\bar{j}\hat{n}} A_{\hat{n}\bar{j}} = \sum_{\hat{n}} \sum_{\bar{j}} B_{\hat{n}\bar{j}} A_{\bar{j}\hat{n}}$, $\text{trace}(AB) = \text{trace}(BA)$

Q4. $C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$

We know that:

inverse of C exists iff $\det(C) \neq 0$.

$\det(C) = 4 \times 6 - 2 \times 7 = 24 - 14 = 10 \neq 0$ Thus, inverse of C exists.

$$\Rightarrow C^{-1} = \frac{1}{4 \cdot 6 - 2 \cdot 7} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

Part 2. Probability, MLE, and MAP Estimation

Q1. $P(X=k) = c \cdot \frac{3^k}{k!}$ $(k=0,1,2,\cdots)$

(a) $\sum_{k=0}^{\infty} c \cdot \frac{3^k}{k!} = c \cdot \sum_{k=0}^{\infty} \frac{3^k}{k!}$

Here, $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{x^k}{k!} \longrightarrow e^3 = \sum_{k=0}^{\infty} \frac{3^k}{k!}$

$\Rightarrow \sum_{k=0}^{\infty} P(X=k) = c \cdot \sum_{k=0}^{\infty} \frac{3^k}{k!} = c \cdot e^3 = 1 \longrightarrow \boxed{c = e^{-3}}$

(b) $E[X] = \sum_{k=0}^{\infty} k \cdot P(X=k)$

$= \sum_{k=0}^{\infty} k \cdot c \cdot \frac{3^k}{k!} = \sum_{k=1}^{\infty} k \cdot c \cdot \frac{3^k}{k!} = 3c \cdot \sum_{k=1}^{\infty} \frac{3^{k-1}}{(k-1)!} = 3c \cdot \sum_{k=0}^{\infty} \frac{3^k}{k!} = 3 \cdot e^{-3} \cdot e^3 = \boxed{3}$

$Var[X] = E[X^2] - E[X]^2$

$= E[X(X-1)+X] - E[X]^2$

$= \sum_{k=1}^{\infty} k(k-1) P(X=k) + E[X] - (E[X])^2$

$= e^{-3} \sum_{k=2}^{\infty} k(k-1) \frac{3^k}{k!} + 3 - 3^2$

$= e^{-3} \sum_{k=2}^{\infty} \frac{9 \cdot 3^{(k-2)}}{(k-2)!} - 6 = 9e^{-3} \sum_{k=0}^{\infty} \frac{3^k}{k!} - 6 = 9e^{-3}e^3 - 6 = 9 - 6 = \boxed{3}$

Q2. $P(A) = 0.5$, $P(B) = 0.3$, $P(A|B) = 0.6$

(a) $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B)}{0.3} = 0.6 \longrightarrow P(A \cap B) = 0.18$

(b) $P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} = \frac{0.6 \cdot 0.3}{0.5} = \frac{0.18}{0.5} = 0.36$

(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.3 - 0.18 = 0.62$

Q3. Assume that every flips are independent.

(a)

$L(\theta) = P(HHTHT|\theta) = \theta^3(1-\theta)^2$

$\rightarrow \log L(\theta) = 3\log\theta + 2\log(1-\theta)$

$\rightarrow \frac{d}{d\theta}\log L(\theta) = \frac{3}{\theta} - \frac{2}{1-\theta} = 0 \rightarrow \frac{3}{\theta} = \frac{2}{1-\theta}$, $3-3\theta = 2\theta$

$\rightarrow 3 = 5\theta$, $\boxed{\theta = \frac{3}{5}}$

(b)

$$P(\theta) = \begin{cases} 2\theta & 0 \le \theta \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Let the event of HHTHT be E.

$\rightarrow P(\theta|E) = \frac{P(E|\theta)\cdot P(\theta)}{P(E)}$

$\rightarrow P(\theta|E) \propto \theta^3(1-\theta)^2 \cdot 2\theta \quad (0 \le \theta \le 1)$

$(= 2\theta^4(1-\theta)^2)$

$\log P(\theta|E) = 4\log\theta + 2\log(1-\theta) + C$, $\frac{d}{d\theta}\log P(\theta|E) = \frac{4}{\theta} - \frac{2}{1-\theta} = 0$

$\rightarrow 4(1-\theta) = 2\theta$, $\boxed{\theta = \frac{2}{3}}$

Q4.

for gaussian with μ, σ^2, $X \sim N(\mu, \sigma^2)$

$\rightarrow P(X = x_i) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$

$\rightarrow L(\mu, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x_i-\mu)^2}{2\sigma^2}\right)$

$\rightarrow \log L(\mu, \sigma^2) = -n\log(\sqrt{2\pi}) - n\log(\sqrt{2\sigma^2}) - \frac{1}{2\sigma^2}\cdot\sum_{i=1}^{n}(x_i-\mu)^2$

$\rightarrow \frac{d}{d\mu}\log L = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i-\mu) = 0 \rightarrow \mu = \frac{1}{n}\sum x_i = \frac{1+3+5+7}{4} = \boxed{4}$

* Note: derivative of σ

Let $s = \sigma^2$

$\rightarrow \frac{\partial l}{\partial s} = -\frac{n}{2s} + \frac{1}{2s^2}\sum_i (x_i-\mu)^2 = 0$

$\rightarrow ns = \sum_i (x_i-\mu)^2$

$\rightarrow \sigma^2 = \frac{1}{n}\sum_i (x_i-\mu)^2 = \frac{1+3+5+7}{4} = \boxed{4}$

$\rightarrow \sigma^2 = \frac{1}{n}\sum(x_i-\mu)^2 = \frac{1}{4}\left((1-4)^2+(3-4)^2+(5-4)^2+(7-4)^2\right) = \frac{1}{4}(9+1+1+9) = \boxed{5}$

Q5. $\bar{x}=5$, $\bar{x}|\mu \sim N(\mu,4)$

prior: $\mu \sim N(0,1)$

by bayes rule, $p(\mu|\bar{x}) \propto p(\bar{x}|\mu)p(\mu)$. $\rightarrow$ likelihood $p(\bar{x}|\mu) \propto \exp\left(-\frac{(\bar{x}-\mu)^2}{2(4)}\right)$; prior: $p(\mu) \propto \exp(-\frac{\mu^2}{2})$

$\Rightarrow \log p(\mu|\bar{x}) = -\frac{(5-\mu)^2}{8} - \frac{\mu^2}{2} + c$

$\frac{d}{d\mu}\log p(\mu|\bar{x}) = \frac{5-\mu}{4} - \mu = 0 \rightarrow \boxed{\mu = 1}$

Q6.

by definition, $\mathrm{Var}(X) = E[(X-E[X])^2]$

$= E[X^2] - 2E[XE[X]] + E[E[X]^2]$

($E[X]$: constant)

$= E[X^2] - 2E[X]E[X] + E[E[X]^2]$

$= E[X^2] - 2(E[X])^2 + (E[X])^2$

$= E[X^2] - (E[X])^2$

Part 3. PCA

Q1. Derive $Var(u^T X) = u^T \Sigma u$

where X: random variable, u: fixed vector

Let $\mu = E[X] \longrightarrow E[u^T X] = u^T E[X] = u^T \mu$.

$\Rightarrow Var[u^T X] = E[(u^T X - E[Y])^2]$ by definition of variance.

$= E[(u^T x - u^T \mu)^2]$

$= E[(u^T (x-\mu)^2)]$

$= E[u^T (x-\mu)(x-\mu)^T u]$) $(u^T(x-\mu))$ is scalar, $u^T(x-\mu) = (x-\mu)^T u$

$= u^T E[(x-\mu)(x-\mu)^T] u$) u is constant

Let covariance matrix $\Sigma = E[(x-\mu)(x-\mu)^T]$

$\longrightarrow Var(u^T X) = u^T \Sigma u$

Q2. Refer to attached code.

Q3. Refer to attached code.

Q4. Refer to attached code.